

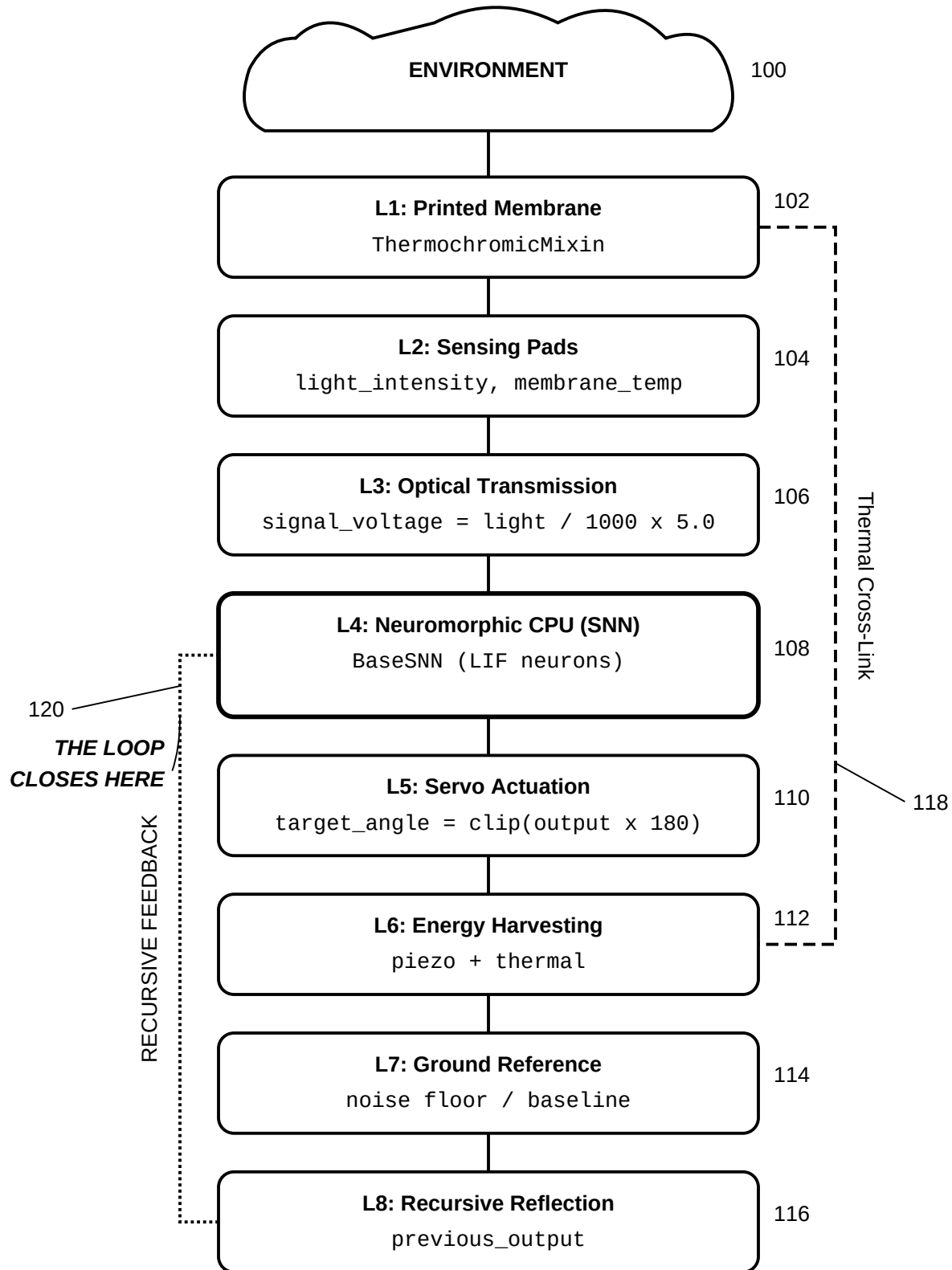


FIG. 1

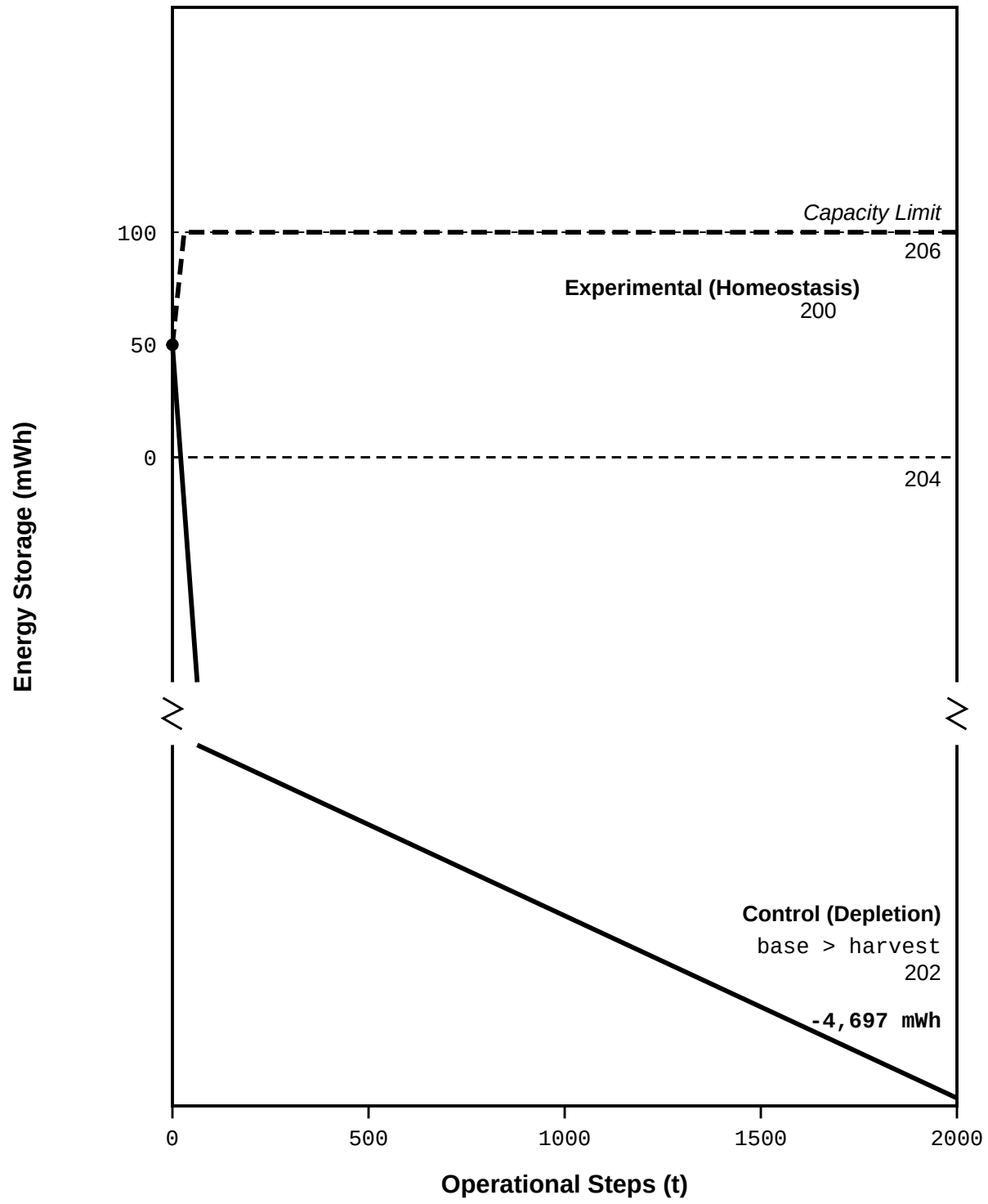
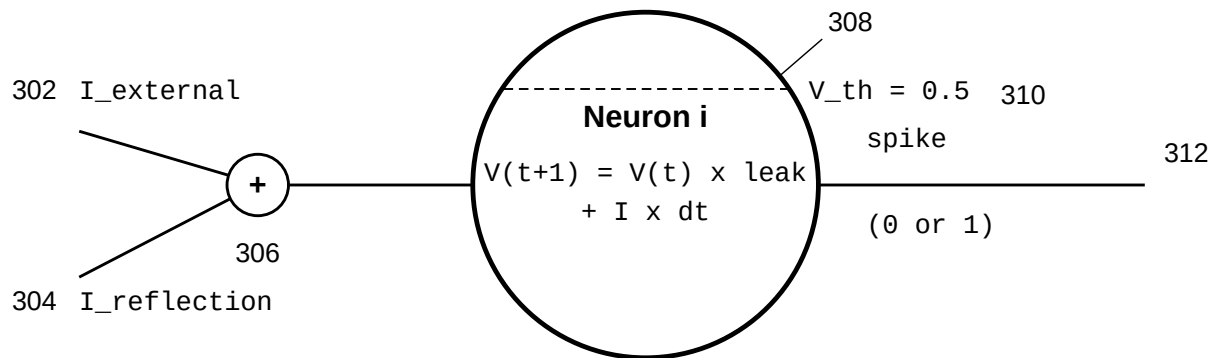


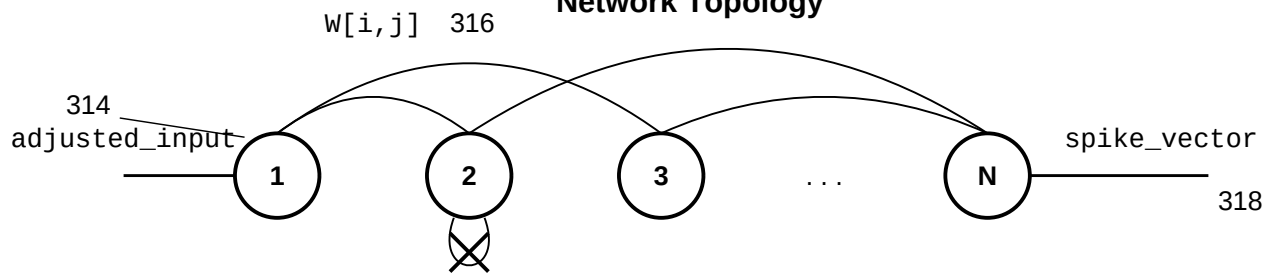
FIG. 2

### Single LIF Neuron Detail



If  $V \geq \text{threshold}$ : spike,  $V = 0$   
 Refractory period: 2 steps

### Network Topology



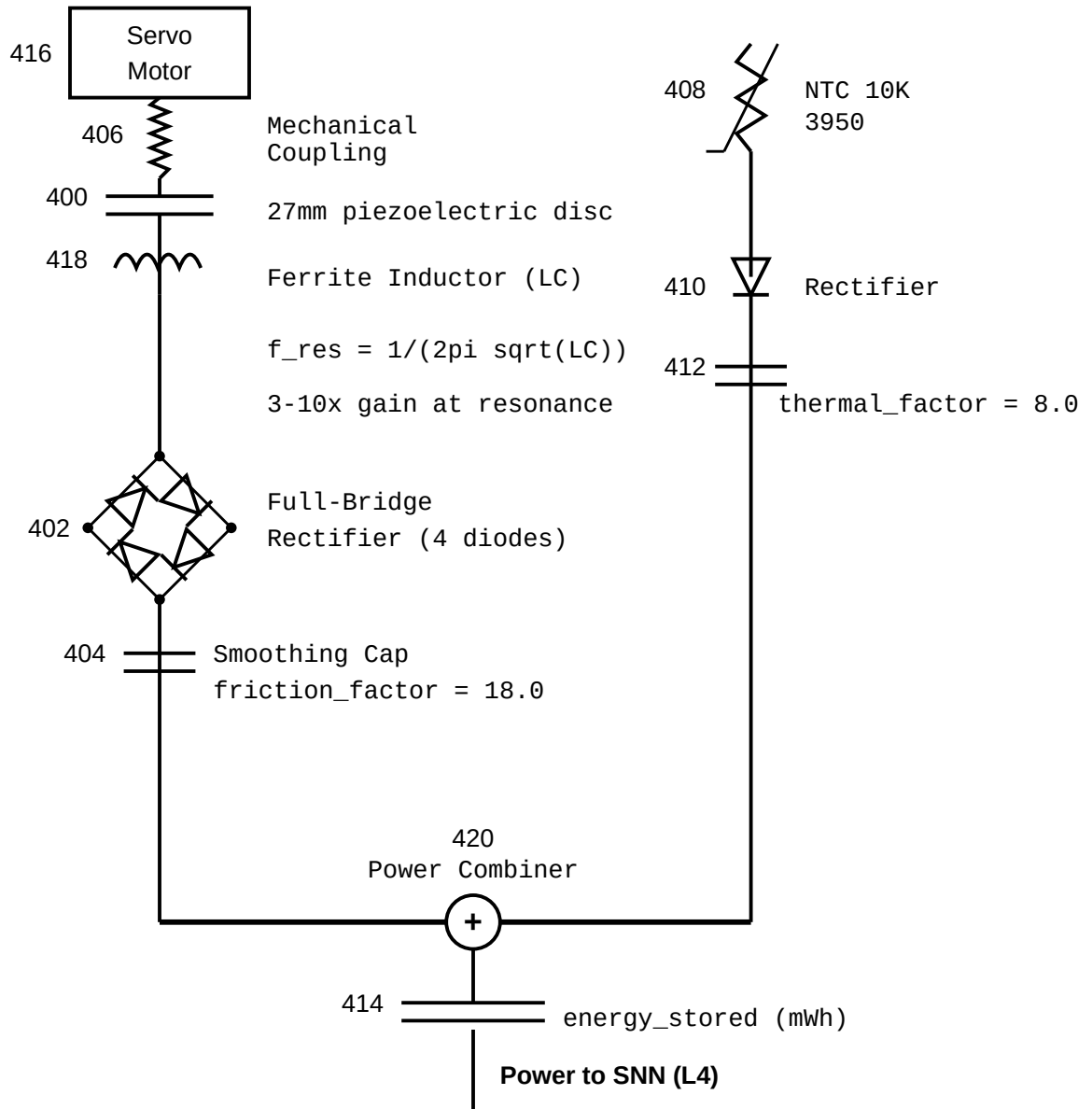
$W[i,i] = 0$  (no self-connections)  
 $N = 50$  (operational),  $N = 20$  (control)

### STDP Learning Rule

Post fires:  $W += A+ \times \text{pre\_trace}$  (LTP)  
 Pre fires:  $W -= A- \times \text{post\_trace}$  (LTD)  
 $A- > A+$  (stability bias)

320

**FIG. 3**

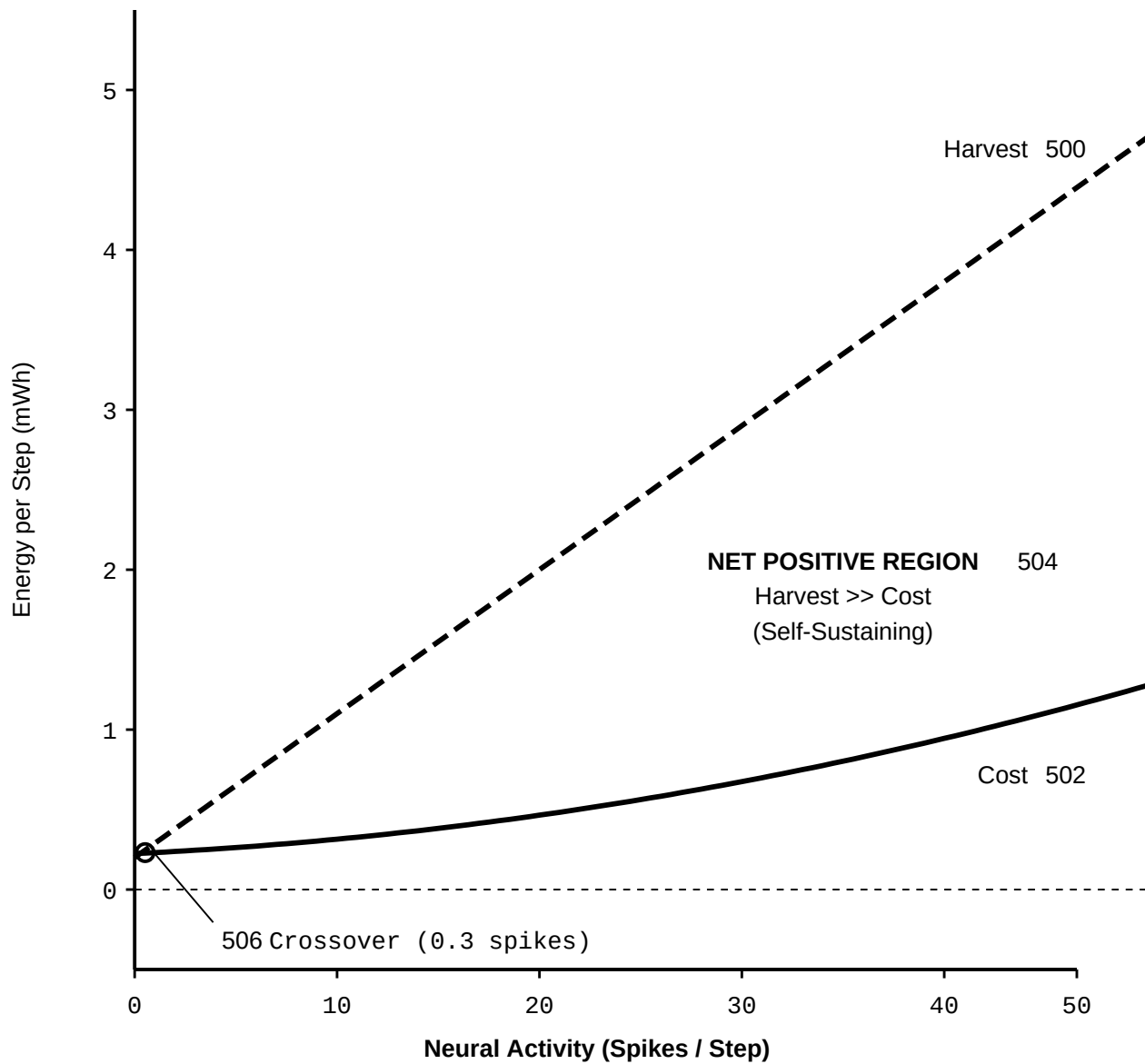
**Piezoelectric Harvester****Thermoelectric Harvester****Power Budget**

Harvest = friction\_harvest + thermal\_harvest

Cost = base\_consumption + activity\_cost x spike\_count<sup>2</sup>

Net = Harvest - Cost

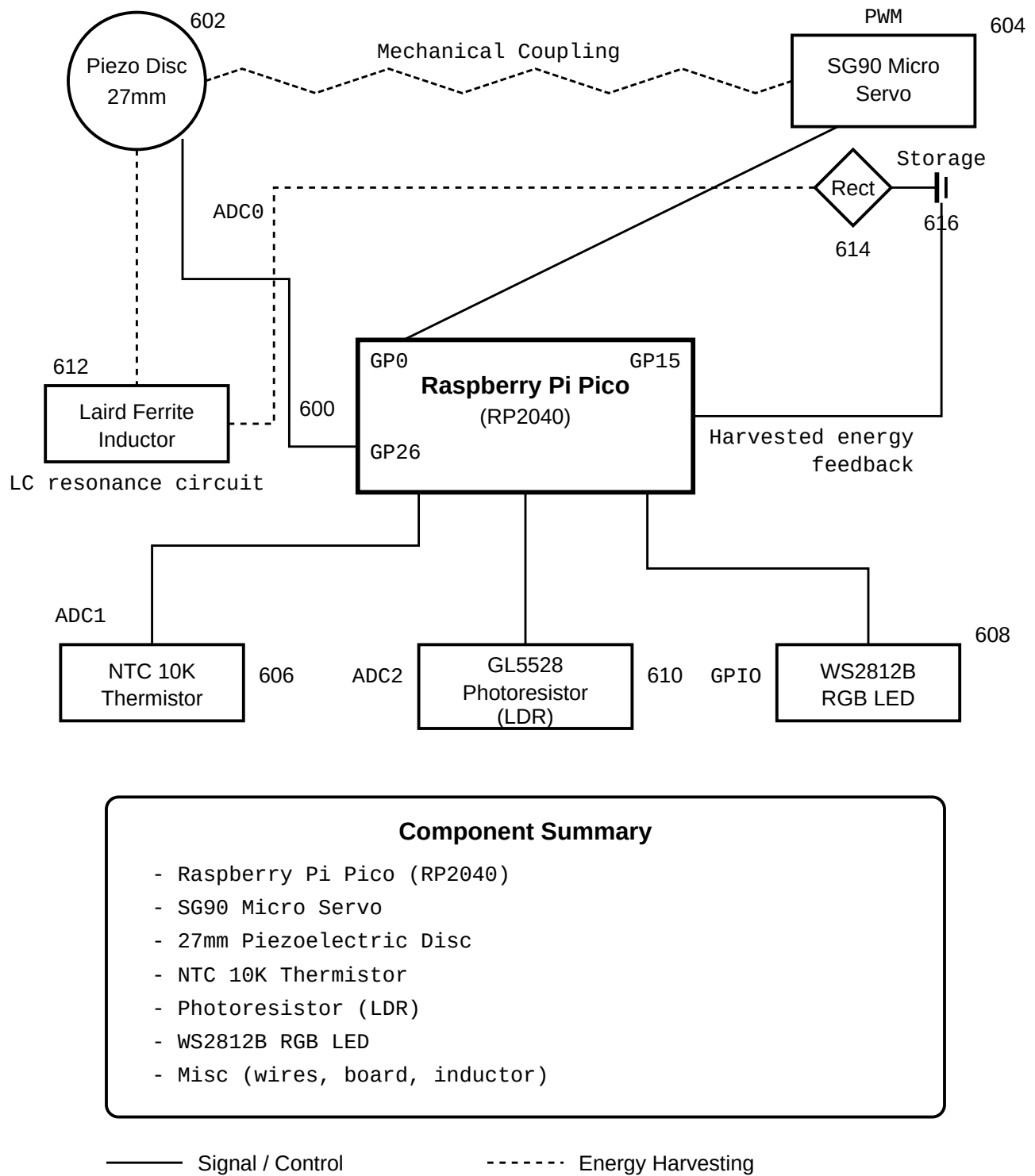
**FIG. 4**



$$\begin{aligned} \text{Harvest} &= 0.20 + (0.09 \times \text{spikes}) \text{ mWh/step} & 508 \\ \text{Cost} &= 0.225 + (0.006 \times s) + (0.0003 \times s^2) \text{ mWh/step} \\ \text{Net} &= \text{Harvest} - \text{Cost} \end{aligned}$$

**Zone 1: Quiescent Drain**  
0 - 1 Spikes: Cost > Harvest

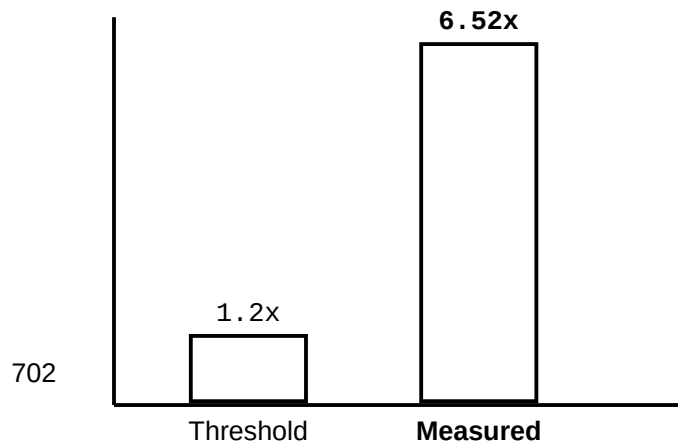
**FIG. 5**

**FIG. 6**

700

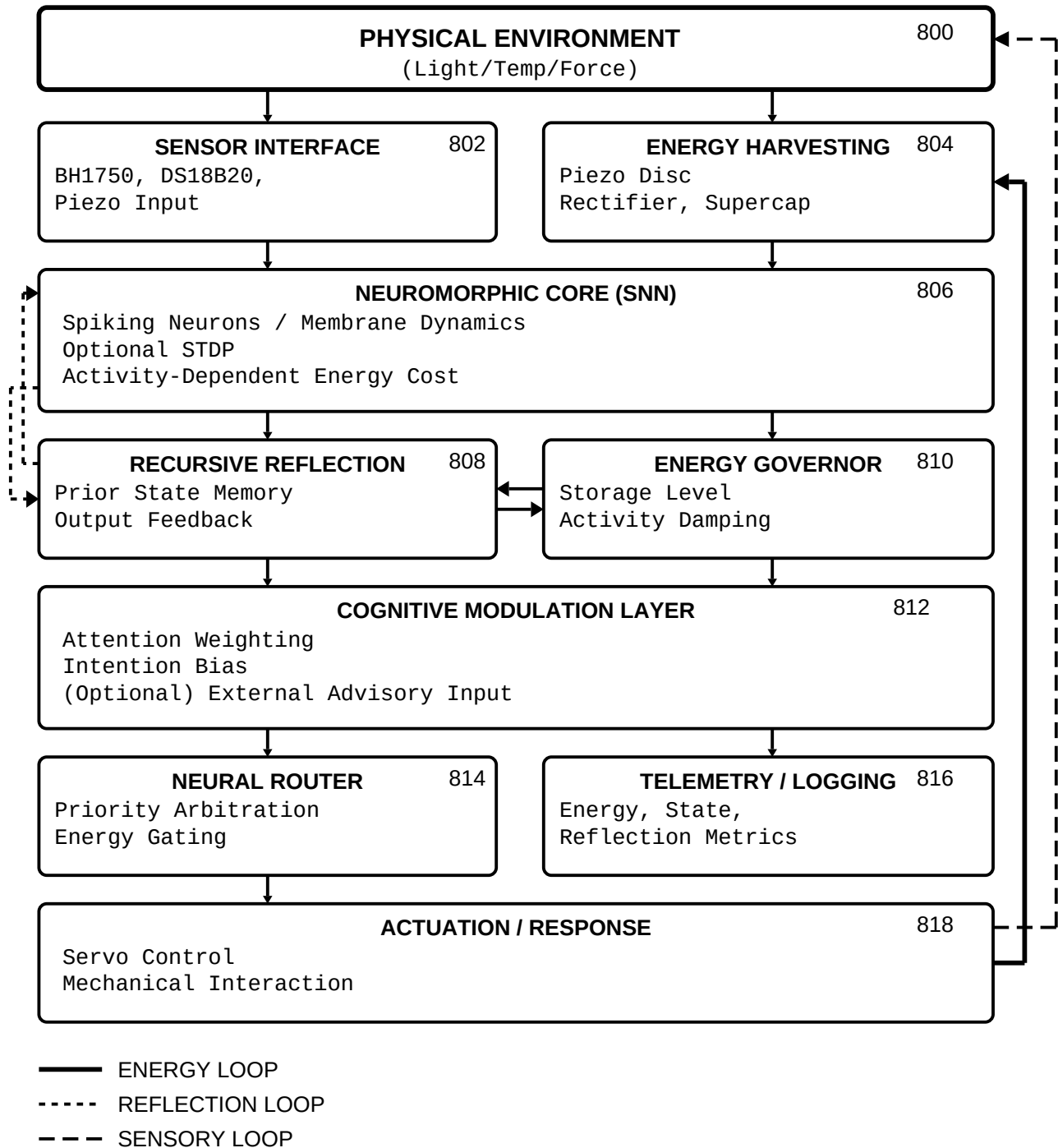
**VALIDATION RESULTS SUMMARY**

| Claim                                                               | Criterion               | Measured           | Threshold | Result           |
|---------------------------------------------------------------------|-------------------------|--------------------|-----------|------------------|
| <b>Phase 7 Control Baseline (EnergyConfig)</b>                      |                         |                    |           |                  |
| A                                                                   | Closed-loop continuity  | 2000 pts           | 2000      | <b>PASS</b>      |
| B                                                                   | Boundedness             | bounded            | bounded   | <b>PASS</b>      |
| C                                                                   | Robustness (noise)      | Std(theta) <= 45   | <= 45     | <b>PASS</b>      |
| D                                                                   | Input-output gain       | corr >= 0.2        | >= 0.2    | <b>PASS</b>      |
| E                                                                   | Saturation ratio        | sat <= 0.20        | <= 0.20   | <b>PASS</b>      |
| <b>Phase 7 Full Integration (Control confirms energy depletion)</b> |                         |                    |           |                  |
| --                                                                  | Energy at 2000 steps    | -4,697 mWh         | < 0       | <b>CONFIRMED</b> |
| <b>MCP Integration (BalancedEnergyConfig)</b>                       |                         |                    |           |                  |
| --                                                                  | Energy at 2000 steps    | 100.0 mWh (capped) | > 0       | <b>CONFIRMED</b> |
| <b>Phase 8 STDP</b>                                                 |                         |                    |           |                  |
| F8.1                                                                | Weight entropy decrease | 2.95 to 2.02 bits  | decrease  | <b>PASS</b>      |
| F8.2                                                                | STDP MI > 1.2x frozen   | 6.52x              | > 1.20    | <b>PASS</b>      |
| F8.3                                                                | Weight convergence      | 0.00%              | < 10%     | <b>PASS</b>      |

**STDP Learning Efficacy (Claim F8.2)**

All claims validated. System is reproducible via git tags.

**FIG. 7**



All computation is constrained by real-time energy availability derived from mechanical interaction.

**FIG. 8**